Midterm

MATH 518 - Fall 2025

October 22, 2025

Total of points = 40 points

Throughout the exam, the field k is an algebraically closed field of characteristic zero. Recall that we define an affine variety to be an *irreducible* algebraic set.

1. (10 points) In the following two cases, describe the algebraic set V, determine the ideal $I(V)$ and if V is irreducible .
   (a) The affine algebraic set $V = V(x^2y^3) \subset k^2$.
   (b) The projective algebraic set $V = V(f, g) \subset \mathbb{P}^2(k)$, where $f = x^2 + y^2 - z^2$ and $g = x - z$ in $k[x, y, z]$.

Solution:

(a) We have $V = V(x^2) \cup V(y^3) = V(x) \cup V(y)$, which shows that it is reducible. It is the union of the yz-plane and the xz-plane. We have $V = V(xy)$ and (xy) is a radical ideal (if xy divides f^n, then xy divides f). Thus $I(V) = (xy)$ by the Nullstellensatz.

(b) We have

$$V(x^2 + y^2 - z^2, x - z) = V(y, x - z) = \{[1 : 0 : 1]\},$$

so V is a unique point. The ideal $I = (y, x - z)$ is prime since

$$k[x, y, z]/(y, x - z) \simeq k[x]$$

is an integral domain. In particular, it is radical, and by the projective Nullstellensatz (V is nonempty)

$$I(V(y, x - z)) = \mathrm{rad}(y, x - z) = (y, x - z)$$

and V is irreducible (which is also follows from the fact that V is a point).

2. (10 points) Find the decomposition of $V = V(y^2 - xz, z^2 - y^3) \subset k^3$ into irreducibles.

Solution: We have

$$\begin{aligned} V(y^2 - xz, z^2 - y^3) &= V(y^2 - xz, z^2 - xyz) \\ &= V(y^2 - xz, z(z - xy)) \\ &= V(y^2 - xz, z) \cup V(y^2 - xz, z - xy) \\ &= V(y, z) \cup V(y(y - x^2), z - xy) \\ &= V(y, z) \cup V(y, z - xy) \cup V(y - x^2, z - xy) \\ &= V(y, z) \cup V(y - x^2, z - x^3). \end{aligned}$$

$V(y, z)$ is clearly irreducible (it is the x-axis), and the second component is irreducible since

$$k[x, y, z]/(y - x^2, z - x^3) \simeq k[x, y]/(y - x^2) \simeq k[x]$$

is an integral domain.

3. Let $\varphi\colon V \longrightarrow W$ be a morphism between two affine varieties $V \subset k^n$ and $W \subset k^m$, and $\varphi^*\colon k[W] \longrightarrow k[V]$ the induced k-algebra morphism.
   (a) (5 points) Show that $I(\varphi(V)) = \ker(\varphi^*)$.
   (b) (5 points) Deduce that φ^* is injective, if and only if $\varphi(V)$ is dense in W. *Such a map is called a dominant map.*

Solution:

(a) *Note that here we identify $I(\varphi(V))$ with its image in $k[W]$, by reducing modulo $I(W)$.* Suppose $\bar{f} \in I(\varphi(V))$. Then for all $p \in V$

$$\varphi^*(\bar{f})(p) = \bar{f}(\varphi(p)) = 0.$$

Thus $\varphi^*(\bar{f}) = 0$. Conversely, if $\bar{f} \in \ker(\varphi^*)$ then $\varphi^*(\bar{f}) = 0$. For all $q = \varphi(p) \in \varphi(V)$ we have

$$\bar{f}(q) = \bar{f}(\varphi(p)) = \varphi^*(\bar{f})(q) = 0.$$

This shows that $I(\varphi(V)) = \ker(\varphi^*)$.

(b) Recall from Homework 1 that the closure of a set $S \subset k^m$ is $\overline{S} = V(I(S))$. Let $\phi\colon k^n \longrightarrow k^m$ be a polynomial map, such that the restriction of ϕ to V gives φ. Then

$$\overline{\varphi(V)} = V(\ker(\phi^*)), \tag{1}$$

where $\phi^*\colon k[x_1,\dots,x_m] \longrightarrow k[x_1,\dots,k_n]$ is the k-algebra morphism that induces φ^* (by passing to the quotient).

If φ^* is injective, then $\ker(\varphi^*) = 0$, or equivalently $\ker(\phi^*) \subset I(W)$. It follows that

$$\overline{\varphi(V)} = V(\ker(\phi^*)) \supset V(I(W)) = W. \tag{2}$$

Since the closure is the smallest closed subset containg $\varphi(V)$, it follows that $\overline{\varphi(V)} = W$, which means that $\varphi(V)$ is dense in W. Conversely, if $V(\ker(\phi^*)) = \overline{\varphi(V)} = W$, then by the Nullstellensatz $\mathrm{rad}(\ker(\phi^*)) = I(W)$. Since any ideal is contained in its radical, this shows $\ker(\phi^*) \subset I(W)$, and so φ^* is injective.

4. (10 points) Show that the affine line k and the hyperbola $V = V(xy-1) \subset k^2$ are not ismorphic as affine varieties.

Solution: By definition, a morphism Φ is of the form

$$\begin{aligned} \Phi\colon k &\longrightarrow V \subset k^2 \\ t &\longmapsto (f(t), g(t)) \end{aligned}$$

for two polynomials $f, g \in k[t]$, and should satisfy $f(t)g(t) = 1$ for all $t \in k$. Thus f and g must be nonzero polynomials of degree 0, and Φ must be a constant function $\Phi(t) = (a, b)$ for all $t \in k$. So it cannot be an isomorphism since it is not even bijective.

Solution: Alternatively, one can use the fact that two varieties are isomorphic if and only if their coordinate rings are isomorphic, as k-algebras. Let Φ^* be the induced k-algebra morphism

$$\Phi^*\colon k[x,y]/(xy-1) = k[\bar{x},\bar{y}] \longrightarrow k[t],$$

where $\bar{x}, \bar{y}$ are the residues of x, y modulo $xy-1$. For any such morphism we have

$$1 = \Phi^*(\bar{x}\bar{y}) = \Phi^*(\bar{x})\Phi^*(\bar{y}), \tag{3}$$

in particular $\Phi^*(\bar{x})$ and $\Phi^*(\bar{y})$ are nonzero units in $k[t]$. This shows that the image of Φ^* lies in the subring $k^\times \subset k[t]$ of nonzero constants, and is not surjective.